A hypocycloid spectral bound for circulant unistochastic matrices without symmetric amplitudes

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Circulant transition probabilities can arise from a unitary matrix whose amplitudes do not share the translation symmetry. We prove that this symmetry mismatch does not enlarge the possible value of any individual Fourier mode. For every dimension d, if a circulant bistochastic matrix $B_{ij} = b_{i-j}$ has a same-dimensional unitary realization $B_{ij} = |U_{ij}|^2$, then each Fourier eigenvalue of B is the normalized trace of the special-unitary multiplicative commutator $D_k U D_l^\dagger U^\dagger$. It therefore lies in the closed unit d-hypocycloid, even when U is not circulant. At $d = 5$ we apply this result to an exact rational example. The circulant matrix generated by $b = (209/500, 3/40, 7/500, 3/40, 209/500)$ satisfies every row and column bracelet inequality, yet its first Fourier eigenvalue lies strictly outside the five-hypocycloid; hence the matrix is not unistochastic. The result shows both that bracelet conditions are insufficient on the five-dimensional circulant slice and that a marginal hypocycloid violation cannot separate arbitrary unitary realizations from translation-invariant unitary realizations. The corresponding five-site phase-recovery conjecture remains unresolved.

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I. INTRODUCTION

A bistochastic matrix describes a finite classical transition law, whereas a unistochastic matrix has the more restrictive representation $B_{ij} = |U_{ij}|^2$ for a unitary matrix U. The latter class connects classical transition probabilities to coherent finite-dimensional dynamics. Its geometry is substantially less transparent than that of the Birkhoff polytope, and even useful necessary conditions need not be sufficient. Rajchel-Mieldzioć *et al.* introduced bracelet matrices as a tractable superset of the unistochastic matrices and studied in detail the intersection with circulant matrices.[1]

Their analysis distinguishes two notions of symmetry. A transition matrix can be circulant even when the unitary that realizes it is not. If the realizing unitary is itself circulant, Ref. 1 proves that every eigenvalue of the transition matrix belongs to the closed unit d-hypocycloid. This region is also the normalized-trace figure of SU(d).[2] Hypocycloids have appeared more generally in the spectral study of unistochastic matrices,[3] while block-valued extensions lead to a distinct generalized notion of unistochasticity.[4]

Here we remove the amplitude-level symmetry assumption from the marginal spectral bound. The key observation is a commutator-trace identity: on the circulant slice, every spatial Fourier eigenvalue is the normalized trace of a multiplicative commutator in SU(d). This produces two consequences. First, a unitary realization whose amplitudes break translation symmetry still obeys exactly the same one-mode hypocycloid constraint. Second, a rational five-dimensional point gives an exact separation between bracelet and unistochastic matrices on the circulant slice.

The result has a deliberately limited scope. It does not decide whether every circulant unistochastic matrix has a circulant unitary realization, a question conjectured affirmatively in Ref. 1. Instead, it proves that any counterexample must evade detection by each marginal hypocycloid test and must involve a genuinely joint phase constraint.

II. CIRCULANT TRANSITION LAWS AND TRACE FIGURES

Fix an integer $d \geq 2$ and label sites by $\mathbb{Z}_d$. Let $\zeta = \exp(2\pi i/d)$ and define a circulant bistochastic matrix by

$$B_{ij} = b_{i-j}, \qquad b_r \geq 0, \qquad \sum_{r=0}^{d-1} b_r = 1, \tag{1}$$

where all indices are taken modulo d. With the Fourier convention in Eq. (1), its eigenvalues are

$$\lambda_k(B) = \sum_{r=0}^{d-1} b_r \zeta^{-kr}, \qquad k = 0, \ldots, d-1. \tag{2}$$

The matrix is unistochastic if there exists $U \in \mathrm{U}(d)$ such that $B_{ij} = |U_{ij}|^2$ for every pair of sites. No translation symmetry is assumed for U.

We denote by $\mathcal{H}_d$ the normalized trace figure

$$\mathcal{H}_d = \left\{ \frac{1}{d} \operatorname{tr} W : W \in \mathrm{SU}(d) \right\}. \tag{3}$$

It is the closed filled unit d-hypocycloid.[2] A boundary parametrization is

$$h_d(\theta) = \frac{d-1}{d} e^{i\theta} + \frac{1}{d} e^{-i(d-1)\theta}. \tag{4}$$

For later use, recall the bracelet condition. For every pair of rows $p \neq q$, the d nonnegative lengths

$$\ell_j = \sqrt{B_{pj} B_{qj}} \tag{5}$$

must obey the polygon inequality $2 \max_j \ell_j \leq \sum_j \ell_j$, and the analogous condition must hold for every pair of columns. These inequalities are necessary for unistochasticity: they express the possibility that the terms in the orthogonality relation between two rows or columns close to form a polygon in the complex plane.[1]

III. A COMMUTATOR-TRACE IDENTITY

Proposition 1. *Let B be a $d \times d$ circulant bistochastic matrix of the form Eq. (1). If $B_{ij} = |U_{ij}|^2$ for an arbitrary $U \in \mathrm{U}(d)$, then, for every k,*

$$\lambda_k(B) = \frac{1}{d} \operatorname{tr} \left(D_k U D_k^\dagger U^\dagger \right) \in \mathcal{H}_d, \qquad D_k = \operatorname{diag}\left(\zeta^{-ki}\right)_{i=0}^{d-1}. \tag{6}$$

Proof. Set $C_k = D_k U D_k^\dagger U^\dagger$. This matrix is unitary, and its determinant cancels multiplicatively:

$$\det C_k = (\det D_k)(\det U)(\det D_k)^*(\det U)^* = 1. \tag{7}$$

Thus $C_k \in \mathrm{SU}(d)$. Expanding its trace and using the fixed displacement convention gives

$$\begin{aligned} \frac{1}{d}\operatorname{tr} C_k &= \frac{1}{d}\sum_{i,j=0}^{d-1} \zeta^{-k(i-j)}|U_{ij}|^2 \\ &= \frac{1}{d}\sum_{r=0}^{d-1}\sum_{i-j=r} \zeta^{-kr} b_r \\ &= \sum_{r=0}^{d-1} b_r \zeta^{-kr} = \lambda_k(B). \end{aligned} \tag{8}$$

There are exactly d ordered pairs (i, j) for each displacement r, which accounts for the cancellation of the prefactor. Equation (3) now implies $\lambda_k(B) \in \mathcal{H}_d$. □

The proof uses only the symmetry of the squared moduli. In particular, it does not assume $[U, S] = 0$ for the one-site translation S, nor does it change dimension or introduce an ancilla. The quantities $\lambda_k(B)$ are the dimensionless one-step Fourier modes of the site-transition law; the identity does not imply that two unitary channels agreeing on these site probabilities agree on coherent inputs.

Relative to the circulant-unitary result of Ref. 1, the proposition enlarges the premise from circulant realizing unitaries to all same-dimensional realizing unitaries, provided the resulting transition matrix is circulant. The conclusion remains a necessary one-mode constraint, not a sufficient condition for either type of realization.

IV. AN EXACT FIVE-DIMENSIONAL BRACELET WITNESS

Consider the rational probability vector

$$b = \left(\frac{209}{500}, \frac{3}{40}, \frac{7}{500}, \frac{3}{40}, \frac{209}{500}\right) \tag{9}$$

and let $B_{ij} = b_{i-j}$ for $i, j \in \mathbb{Z}_5$.

Corollary 1. *The matrix defined by Eq. (9) is circulant, bistochastic, and satisfies every row and column bracelet inequality, but it is not unistochastic.*

Proof. Normalization and nonnegativity are immediate. Write $p = 209/500$, $q = 3/40$, and $r = 7/500$. For a row separation of one site, the bracelet lengths, up to cyclic order, are

$$\sqrt{pq},\ \sqrt{qr},\ \sqrt{qr},\ \sqrt{pq},\ p. \tag{10}$$

The only nonautomatic polygon inequality has exact margin

$$m_1 = -\frac{209}{500} + \frac{\sqrt{42} + \sqrt{1254}}{100}. \tag{11}$$

The integer bounds $31350 > 177^2$ and $1050 > (162/5)^2$ give $m_1 > 1/1250 > 0$. For a separation of two sites, the length multiset is

$$\sqrt{pr},\ q,\ \sqrt{pr},\ \sqrt{pq},\ \sqrt{pq}. \tag{12}$$

Its largest member $\sqrt{pq}$ occurs twice, so the polygon inequality follows immediately. Separations three and four reproduce these two multisets, and the column tests are identical by cyclic reindexing. Hence all bracelet conditions hold.

For the first Fourier mode, exact cyclotomic reduction yields

$$\lambda_1(B) = te^{i\pi/5}, \qquad t = \frac{465 + 343\sqrt{5}}{2000}. \tag{13}$$

The radial endpoint of the filled five-hypocycloid on this symmetry ray is

$$h_5(\pi/5) = \frac{3}{5}e^{i\pi/5}. \tag{14}$$

Since $\sqrt{5} > 15/7$, Eq. (13) gives $t > 3/5$ strictly. Thus $\lambda_1(B) \notin \mathcal{H}_5$. The proposition rules out every same-dimensional unitary realization, proving that B is not unistochastic. □

This is not a counterexample to the phase-recovery conjecture: the witness fails its essential premise because it has no arbitrary unitary realization. Instead, it gives a five-dimensional negative answer to the narrower question whether all bracelet conditions suffice for unistochasticity on the circulant slice.

V. CONSEQUENCE FOR TRANSLATION-SYMMETRIC REALIZATION

Let $\mathcal{C}_d$ denote the circulant bistochastic matrices, $\mathcal{U}_d$ the unistochastic matrices, and $\mathcal{C}_d^2$ the circulant matrices that admit a circulant unitary realization, following Ref. 1. That work proves

$$\mathcal{C}_d \cap \mathcal{U}_d = \mathcal{C}_d^2 \qquad (d \le 4) \tag{15}$$

and conjectures the equality in every dimension. The first unsettled case in that program is

$$\mathcal{C}_5 \cap \mathcal{U}_5 \stackrel{?}{=} \mathcal{C}_5^2. \tag{16}$$

The proposition shows that every element on the left-hand side of Eq. (16) already satisfies every marginal five-hypocycloid constraint. The same constraints were previously known for the right-hand side. Therefore an eigenvalue outside $\mathcal{H}_5$ can reject a circulant bistochastic or bracelet candidate, as in the corollary, but it can never distinguish the two realization classes in Eq. (16). If the equality fails, its separator must retain information about the joint compatibility of the nontrivial Fourier modes rather than test them independently.

Physically, the result is a one-step control limitation on a fixed finite ring. Even when the amplitude matrix breaks translation symmetry, a translation-invariant table of all site-to-site probabilities cannot push an individual spatial Fourier mode beyond the same trace figure obeyed by a translation-invariant amplitude matrix. This statement is exact and dimensionless. It invokes no perturbative order, continuum or thermodynamic limit, coin degree of freedom, environment, mixture, or postselection.

VI. METHODS AND REPRODUCIBILITY

The proof of Eq. (6) is an exact index calculation. The five-dimensional certificate uses rational arithmetic, symbolic radicals, and strict integer inequalities; no fitted parameter, optimizer, sampling argument, or floating-point tolerance enters the conclusion. A deterministic SymPy script supplied with the accompanying reproducibility package checks the normalization, every bracelet class, Eq. (13), and the strict hypocycloid exclusion. It also tests the sign convention on an exact unitary whose squared moduli are circulant although the unitary does not commute with translation.

OpenAI Codex (OpenAI; model reported by the research environment as GPT-5, session date September 13, 2026) was used to assist with derivation, exact symbolic checking, bounded literature screening, and manuscript drafting. Its outputs are not treated as independent experimental data or external peer review. The named author must critically review the calculations, references, scope, and wording and assume responsibility for any submitted version.

VII. CONCLUSION

Every Fourier eigenvalue of a circulant unistochastic matrix is a normalized special-unitary commutator trace, whether or not the realizing unitary is circulant. The resulting all-d hypocycloid bound turns an explicit rational five-site bracelet matrix into an exact nonunistochastic witness and proves that marginal spectral exclusion cannot resolve the distinction between arbitrary and translation-symmetric unitary realizations. The full five-site phase-recovery question in Eq. (16) remains unresolved; the next structural target is the joint compatibility of its nontrivial Fourier modes.

AUTHOR DECLARATIONS

Conflict of interest

The conflict-of-interest declaration must be confirmed by the author before submission.

Author contributions

The CRediT contribution statement must be confirmed by the author before submission.

Data availability

No empirical data were generated. The exact symbolic verification script, software requirements, regeneration instructions, and reference output are included in the accompanying reproducibility package.

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